R110 | Whole-version A/B comparison

Same frozen data, common base and prototype initialization; NOT parameter/FLOP-matched.

Split	Arm	Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	regret (%)
TRAIN	A	71.616941	11.388930	11.400702	11.345073	-0.198522	0.386376	0.395426
TRAIN	B	71.264472	11.389609	11.400702	11.345073	-0.193596	0.391332	0.399642
VAL	A	71.103704	11.416779	11.427978	11.371692	-0.192984	0.394679	0.405093
VAL	B	70.896296	11.417566	11.427978	11.371692	-0.184524	0.403215	0.412389
TEST	A	71.307407	11.374707	11.385299	11.329932	-0.186717	0.396557	0.406464
TEST	B	71.233333	11.374359	11.385299	11.329932	-0.192514	0.390729	0.399449

A_tied_ce | parameters=2,305,033 | selected update=43200

SHA256: 1faa6a4190cd3cda9d00e6816b0d88f19e0730a6284190450f0872587ddf5ff0

B_private_ce | parameters=10,220,553 | selected update=21600

SHA256: 34c0d7575e8ccad9cfa5a19849d2d8c404f63b2244dca688bc4bcd40f22cd59b

R110 refines each candidate node graph through two layers on shared original H4 before pooling. A_tied_ce ties these weights; B_private_ce has 21 banks keyed by global method ID, never candidate column, winner or outcome. Both retain candidate conditioning, original two joint layers and score MLP/maximin. TRAIN optimizes one-forward unweighted legal-pool raw-utility CE, weight 1; comparison, relative-risk and RDrop/KL weights are zero. Evaluation maximin CE is diagnostic, not the optimized raw-utility CE. A has 2,305,033 trainables; B has 10,220,553. The original 1,842,953-parameter base is fresh matched, and all late branches start from identical original-layer prototypes. The initial eval function is identical, but complete state hashes/dimensions are not identical. This is not parameter/FLOP-matched, a routing experiment, or an isolated capacity-only control. Original R88B physical TSP node0 role, methodID+34 and geometry/distance remain; 57,600 updates, logical batch1024, seed2. No new route/path fields, labels, donor, teacher, context, targets, calibration, quotas or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B, R93A and R109 arms are explicit optional saved references, never implicit dependencies. NSS uses saved historical R79 weights, not retrained on new90k. Fixed36 true rescues/losses, false picks and aligned population net costs/regret are separate from selection frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint per arm for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete method-conditioned node refinement, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. Microcaps/eval batch come from the main-witness locked runner, not this reporter. No routing summary is required. No online W&B equality/availability is asserted.

R110 | TRAIN: ALL and18 tasks

Native/FP64 ind; same per-arm lock; mean cost and actual regret remain distinct.

Problem	A Top1 (%)	B Top1 (%)	A mean_cost	B mean_cost	A regret (%)	B regret (%)	B-A cost
ALL	71.6169	71.2645	11.3889301	11.3896091	0.39543	0.39964	0.0006790
TSP	46.3494	46.3593	8.0045548	8.0049674	0.42688	0.43235	0.0004126
CVRP	48.3025	46.3247	18.0427231	18.0456192	0.71215	0.70402	0.0028961
ATSP	76.7049	76.4716	1.6403389	1.6403200	0.39153	0.39661	-0.0000189
OVRP	42.0864	41.8148	7.4432796	7.4437421	0.62865	0.63494	0.0004624
VRPB	80.9580	80.6358	9.3573937	9.3561069	0.41701	0.40200	-0.0012869
VRPL	38.7321	37.7654	12.3920743	12.3938383	0.39139	0.40542	0.0017640
VRPTW	80.5469	80.4531	18.8118045	18.8125292	0.30098	0.30511	0.0007247
OVRPTW	83.9704	83.9136	10.1180860	10.1180895	0.24091	0.24164	0.0000034
OVRPB	88.4988	88.4333	6.3632688	6.3633045	0.26804	0.27101	0.0000357
OVRPL	42.9506	41.5864	7.4432232	7.4449604	0.61495	0.63871	0.0017372
VRPBL	80.2062	79.8802	9.4021904	9.4017612	0.41734	0.41385	-0.0004292
VRPBTW	79.7630	79.6815	19.7301677	19.7317657	0.42964	0.43761	0.0015980
VRPLTW	80.6432	80.5296	18.7675042	18.7686359	0.29809	0.30465	0.0011317
OVRPBL	88.6815	88.6111	6.3587544	6.3591616	0.26374	0.27103	0.0004072
OVRPBTW	83.4086	83.3457	10.6233121	10.6235133	0.32425	0.32664	0.0002013
OVRPLTW	83.9383	83.8247	10.1102659	10.1106425	0.23935	0.24354	0.0003766
VRPBLTW	79.9025	79.7358	19.7672179	19.7690746	0.42472	0.43323	0.0018567
OVRPBLTW	83.4617	83.3938	10.6245817	10.6249308	0.32803	0.33119	0.0003492

ALL task-equal; raw costs across different problems are descriptive, not a common-unit objective.

SBS fitted on frozen TRAIN; Oracle is each instance legal-pool minimum. No TEST/LIB reselection.

R110 | VAL: ALL and18 tasks

Native/FP64 ind; same per-arm lock; mean cost and actual regret remain distinct.

Problem	A Top1 (%)	B Top1 (%)	A mean_cost	B mean_cost	A regret (%)	B regret (%)	B-A cost
ALL	71.1037	70.8963	11.4167790	11.4175660	0.40509	0.41239	0.0007870
TSP	46.8889	46.2000	8.0255318	8.0261143	0.42687	0.43729	0.0005825
CVRP	46.8667	44.9333	18.1658010	18.1657039	0.75369	0.73263	-0.0000971
ATSP	76.7556	76.5556	1.6388384	1.6392884	0.38669	0.41934	0.0004500
OVRP	40.9556	41.0444	7.4613501	7.4632541	0.62522	0.65107	0.0019040
VRPB	81.1778	81.2000	9.3548867	9.3552241	0.39141	0.39132	0.0003374
VRPL	37.2444	37.1111	12.4307550	12.4325866	0.38922	0.40346	0.0018316
VRPTW	80.0889	79.9556	18.8187670	18.8196785	0.29580	0.30181	0.0009115
OVRPTW	83.8889	83.8667	10.1430275	10.1425368	0.25783	0.25058	-0.0004907
OVRPB	88.6444	88.5111	6.3582745	6.3586527	0.28180	0.28877	0.0003782
OVRPL	41.6222	41.0444	7.4379437	7.4392395	0.65917	0.67527	0.0012958
VRPBL	78.9556	78.8889	9.3686201	9.3686110	0.43556	0.43328	-0.0000091
VRPBTW	78.4889	78.6667	19.9464169	19.9492476	0.46301	0.46712	0.0028308
VRPLTW	78.8444	78.9556	18.7639720	18.7645341	0.31512	0.31613	0.0005621
OVRPBL	89.3778	89.3333	6.3686750	6.3698822	0.25930	0.28099	0.0012072
OVRPBTW	83.6889	83.5778	10.6596358	10.6603898	0.31300	0.32226	0.0007539
OVRPLTW	83.5778	83.5556	10.1563436	10.1570707	0.26159	0.27482	0.0007271
VRPBLTW	79.6222	79.4889	19.7924478	19.7942890	0.42533	0.43437	0.0018412
OVRPBLTW	83.1778	83.2444	10.6107358	10.6098852	0.35106	0.34248	-0.0008506

ALL task-equal; raw costs across different problems are descriptive, not a common-unit objective.

SBS fitted on frozen TRAIN; Oracle is each instance legal-pool minimum. No TEST/LIB reselection.

R110 | TEST: ALL and18 tasks

Native/FP64 ind; same per-arm lock; mean cost and actual regret remain distinct.

Problem	A Top1 (%)	B Top1 (%)	A mean_cost	B mean_cost	A regret (%)	B regret (%)	B-A cost
ALL	71.3074	71.2333	11.3747066	11.3743591	0.40646	0.39945	-0.0003475
TSP	47.3111	47.1333	7.9719402	7.9723081	0.42573	0.43031	0.0003678
CVRP	48.6000	46.6444	18.0138027	18.0165874	0.70667	0.70945	0.0027847
ATSP	76.2889	76.5111	1.6360063	1.6357843	0.40282	0.39214	-0.0002220
OVRP	40.4667	41.2000	7.4458137	7.4456004	0.62265	0.62165	-0.0002133
VRPB	80.4889	80.6000	9.3540982	9.3503511	0.42626	0.38001	-0.0037471
VRPL	37.5111	36.9556	12.3494528	12.3516180	0.40285	0.41815	0.0021652
VRPTW	81.1778	81.1778	18.6880007	18.6884197	0.28434	0.28479	0.0004190
OVRPTW	83.9111	83.9111	10.0927295	10.0927135	0.23512	0.23455	-0.0000160
OVRPB	88.0667	88.2000	6.3736350	6.3704309	0.35221	0.29523	-0.0032041
OVRPL	41.4667	41.6667	7.4643904	7.4644302	0.63609	0.63452	0.0000398
VRPBL	79.2222	79.4667	9.3987827	9.3958119	0.45360	0.42592	-0.0029708
VRPBTW	78.9778	78.9111	19.6659379	19.6643978	0.44558	0.44048	-0.0015401
VRPLTW	79.3111	79.1778	18.8039707	18.8045017	0.33071	0.33341	0.0005310
OVRPBL	89.3333	89.2444	6.3755424	6.3758475	0.26776	0.27199	0.0003051
OVRPBTW	85.0222	84.8889	10.6586357	10.6589785	0.29184	0.29401	0.0003428
OVRPLTW	82.9333	83.0667	10.0853940	10.0853319	0.26203	0.26183	-0.0000621
VRPBLTW	79.9111	79.8667	19.7040230	19.7033586	0.44455	0.44239	-0.0006643
OVRPBLTW	83.5333	83.5778	10.6625634	10.6619929	0.32553	0.31925	-0.0005705

ALL task-equal; raw costs across different problems are descriptive, not a common-unit objective.

SBS fitted on frozen TRAIN; Oracle is each instance legal-pool minimum. No TEST/LIB reselection.

R110 | Frozen36 recall and population cost

True rescues/losses and false picks, not selection counts; B versus A.

Split	Winners	A recall (%)	B recall (%)	Rescued	Lost	New false	ALL cost delta	regret delta pp
TRAIN	83279	2.011311	1.036276	389	1201	629	0.000678993	0.004216
VAL	4625	1.945946	0.843243	17	68	36	0.000786992	0.007296
TEST	4598	1.522401	0.956938	21	47	41	-0.000347507	-0.007015

Split	New false cost	Removed false cost	Target winner cost	Full18 net cost
TRAIN	0.000140231	-0.000160526	0.000098370	0.000678993
VAL	0.000117496	-0.000074296	0.000070274	0.000786992
TEST	0.000099192	-0.000301261	0.000083557	-0.000347507

Target36 SHA256: 406ded5a21d8b73c2a41b87dc2bc3cc14330663ae1aa4f59ca6fbfde83770fe4

Target cost contributions are task-equal; fixed-target changes are not full-population net gain.

R89A/R89B/R84A and historical R79 NSS remain saved references in each arm PDF.